

ZAK MCP Gateway v0.4

6 Oct 2026 · Fewer turns, cheaper turns: 7 improvements

Targets what the lean test showed: searches before writes, the fixed cost of every turn, and multi-step build jobs. All options are off by default; defaults behave exactly like v0.3.

In one picture: "The usual, please": the restaurant already knows the steps of the order and runs them all; you get one confirmation.

What's new

Feature	What it does (simply)	Where to switch it
#1 Pinned signatures	Configured tools' one-line signatures always in the instructions, so writes need no search.	<code>discovery.pinnedSignatures</code>
#2 Workflow tools	A configured multi-step job the agent runs in one call (<code>wf__name</code>); values flow between steps; one confirmation; stops and reports at the first failure; output trimmed like normal replies.	<code>gateway.workflows</code>
Workflow suggestions	<code>npm run report</code> finds repeated call sequences and prints a workflow skeleton to review.	<code>npm run report</code>
#3 Cheaper fixed cost per turn	Slim meta-tools; tool-inventory sentences of server instructions not repeated; groups catalog.	<code>discovery.metaToolStyle</code> , <code>catalog</code>
#4 Leaner search replies	Top hit on one line, the rest as names only.	<code>discovery.signatureStyle</code> , <code>alsoStyle</code>
#5 Search re-ranking	Tools whose name words match beat descriptions that repeat a common word; count synonyms.	built in
Multi-table replies	An object holding several lists renders as several small tables.	built in
#6 History trimming (host)	Agent UI shortens old results to handle stubs; recommended for every agent host.	<code>agent-ui</code> switch
#7 Prompt caching (host)	Keep the tool list fixed so the host's prompt cache bills it at ~10%; note: focused tools change it per question.	host setting

Fixed

Problem	Result
Workflow output carried raw step data (found in the GitHub run: 28,384 tokens)	Output uses trimmed views; 808 tokens
Search: 'add a field to a table' ranked the column tool 3rd	Name re-ranking puts it 1st
Several lists in one reply rendered as JSON	Rendered as table sections

Measured

What	Result
GitHub, 9 tasks, agent-run input	none 1,822,172 · v0.3 75,586–101,171 · v0.4 tuned 58,715–80,603
Realistic: v0.3 searching vs v0.4 with pinned signatures	101,171 → 58,715 (–42%), 18 → 11 model requests
Repo overview: 3 calls vs 1 workflow	753 tokens over 3 turns → 808 tokens in 1 turn
Search accuracy	GitHub + filesystem: 32/32 top-3, 31/32 top-1 (unchanged); lean-style catalog: 5/5 top-1

Notes

- Pinned signatures cost ~100 tokens per turn; they pay off only when they remove searches. Pin the 10–12 tools used most, not everything.
- Workflows sit next to the normal tools; give each a clear "use for / not for" description.
- Next: re-run the company lean benchmark with pinned write signatures and one build workflow, then a live-model run.

All versions: docs/versions · Full explanation: docs/ZAK_Gateway_How_It_Works.pdf